Lecture 10: Duality Theory

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Review: Simplex Method

We have completed our discussions on the simplex method.

- The idea of simplex method (search among the BFS and search among the neighbors) and its justifications
- The algebraic procedures
- The simplex tableau
- Other issues about simplex method (initialization, degeneracy, etc.)

The simplex method is not a polynomial time algorithm for any configuration that people have tried.

- However, linear optimization is polynomial-time solvable.
- Later we will discuss another algorithm for LP — the interior point method, but now we turn to introduce another very important theory for linear program, the duality theory.

Duality Theory: Motivation

Consider the standard LP ($m \times n$):

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

We can write it as:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} + \max_{\mathbf{y}\in\mathbb{R}^m} \mathbf{y}^T(\mathbf{b} - A\mathbf{x}) \\ \text{subject to} \quad & \mathbf{x} \geq 0 \end{aligned}$$

Why?

- If $A\mathbf{x} \neq \mathbf{b}$, then we can find $\mathbf{y}$ such that $\mathbf{y}^T(A\mathbf{x} - \mathbf{b}) = \infty$, thus can't be optimal. So we implicitly enforce the constraint $A\mathbf{x} = \mathbf{b}$

Continue..

$$\min_{\mathbf{x}\geq 0} \quad \max_{\mathbf{y}} \quad \mathbf{c}^T\mathbf{x} + \mathbf{y}^T(\mathbf{b} - A\mathbf{x})$$

Now we assume that we can exchange the max and min (will justify it later). Then the problem becomes...

$$\max_{\mathbf{y}} \quad \mathbf{b}^T\mathbf{y} + \min_{\mathbf{x}\geq 0} \mathbf{x}^T(\mathbf{c} - A^T\mathbf{y})$$

We claim that this is equivalent to

$$\begin{aligned} &\text{maximize}_{\mathbf{y}} && \mathbf{b}^T\mathbf{y} \\ &\text{subject to} && A^T\mathbf{y} \leq \mathbf{c} \end{aligned}$$

Why?

$$\min_{\mathbf{x}\geq 0} \mathbf{x}^T(\mathbf{c} - A^T\mathbf{y}) = \begin{cases} 0 & \text{if } A^T\mathbf{y} \leq \mathbf{c} \\ -\infty & \text{if } A^T\mathbf{y} \not\leq \mathbf{c} \end{cases}$$

Now We Get Two Linear Programs...

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

and

$$\begin{aligned} \text{maximize}_{\mathbf{y}} \quad & \mathbf{b}^T\mathbf{y} \\ \text{subject to} \quad & A^T\mathbf{y} \leq \mathbf{c} \end{aligned}$$

We call them dual to each other. If we call one the *primal problem*, then the other one is called *the dual problem*.

- We call $\mathbf{y}$ the dual variables (to the first LP)
- By our derivation, they should have the same optimal value (provided the exchange of min and max is valid)

Dual Problem

Now we have derived our first pair of dual problems.

- Duality theory is very important for linear optimization (and for general optimization problems as well)
- The dual problem carries much useful information for the primal problem
- It also helps to solve optimization problems

Different Duality Forms

We have derived the dual problem of the standard form. What if we want to find the dual problem of:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} \geq \mathbf{b} \end{aligned} \tag{1}$$

Of course, one can first transform (1) to the standard form and then derive the dual. But we can directly apply our previous arguments.

The primal problem can be written as

$$\min_{\mathbf{x}} \quad \mathbf{c}^T\mathbf{x} + \max_{\mathbf{y}\geq 0} \mathbf{y}^T(\mathbf{b} - A\mathbf{x})$$

Why?

$$\max_{\mathbf{y}\geq 0} \mathbf{y}^T(\mathbf{b} - A\mathbf{x}) = \begin{cases} 0 & \text{if } A\mathbf{x} \geq \mathbf{b} \\ \infty & \text{if } A\mathbf{x} \ngeq \mathbf{b} \end{cases}$$

Continue

Assume we can exchange the order of max and min, we get

$$\max_{\mathbf{y}\geq 0} \quad \mathbf{b}^T\mathbf{y} + \min_{\mathbf{x}} \mathbf{x}^T(\mathbf{c} - A^T\mathbf{y})$$

This is further equivalent as

$$\begin{aligned} \text{maximize} \quad & \mathbf{b}^T\mathbf{y} \\ \text{subject to} \quad & A^T\mathbf{y} = \mathbf{c} \\ & \mathbf{y} \geq 0 \end{aligned}$$

We get the dual problem of (1)

Generally

Primal			**Dual**		
minimize	$\mathbf{c}^T\mathbf{x}$		maximize	$\mathbf{b}^T\mathbf{y}$	
subject to	$\mathbf{a}_i^T\mathbf{x} \geq b_i,$	$i \in M_1,$	subject to	$y_i \geq 0,$	$i \in M_1$
	$\mathbf{a}_i^T\mathbf{x} \leq b_i,$	$i \in M_2,$		$y_i \leq 0,$	$i \in M_2$
	$\mathbf{a}_i^T\mathbf{x} = b_i,$	$i \in M_3,$		y_i free,	$i \in M_3$
	$x_j \geq 0,$	$j \in N_1,$		$A_j^T\mathbf{y} \leq c_j,$	$j \in N_1$
	$x_j \leq 0,$	$j \in N_2,$		$A_j^T\mathbf{y} \geq c_j,$	$j \in N_2$
	x_j free,	$j \in N_3,$		$A_j^T\mathbf{y} = c_j,$	$j \in N_3$

- $\mathbf{a}_i^T$ is the ith row of A, A_j is the jth column of A
- Each primal constraint corresponds to a dual variable
- Each primal variable corresponds to a dual constraint
- Equality constraints always correspond to free variables

Example

$$\begin{aligned} \text{minimize} \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \geq 5 \\ & x_1 - x_2 \leq 3 \\ & x_1 \geq 0, x_2 \text{ free} \end{aligned}$$

The dual problem:

- Associate constraint 1 with dual variable y_1
- Associate constraint 2 with dual variable y_2

$$\begin{aligned} \text{maximize} \quad & 5y_1 + 3y_2 \\ \text{s.t.} \quad & y_1 + y_2 \leq 1 \\ & y_1 - y_2 = 2 \\ & y_1 \geq 0, y_2 \leq 0 \end{aligned}$$

To Memorize

Primal	minimize	maximize	Dual
constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 free	variables
variables	≥ 0 ≤ 0 free	$\leq c_j$ $\geq c_j$ $= c_j$	constraints

My trick to memorize:

1. Equality constraints correspond to free variables, vice versa
2. I call non-negative constraints *usual* constraints; non-positive *unusual* constraints. When maximizing, I call $\leq c_i$ *usual* constraint (upper bound for each resource), and $\geq c_i$ *unusual*; when minimizing, I call $\geq b_i$ *usual* and $\leq b_i$ *unusual*.
3. Then usual (unusual) constraints always correspond to usual (unusual) constraints.

Example

Consider the following problem:

$$\begin{array}{lllll} \text{minimize} & x_1 & +2x_2 & +3x_3 & \\ \text{subject to} & -x_1 & +3x_2 & & = 5 \\ & 2x_1 & -x_2 & +3x_3 & \geq 6 \\ & & & x_3 & \leq 4 \\ & x_1 \geq 0 & x_2 \leq 0 & x_3 \text{ free} & \end{array}$$

The dual is:

$$\begin{array}{lllll} \text{maximize} & 5y_1 & +6y_2 & +4y_3 & \\ \text{subject to} & -y_1 & +2y_2 & & \leq 1 \\ & 3y_1 & -y_2 & & \geq 2 \\ & & +3y_2 & +y_3 & = 3 \\ & y_1 \text{ free} & y_2 \geq 0 & y_3 \leq 0 & \end{array}$$

One More Example

Recall the support vector machine problem. The primal problem is:

$$\begin{aligned} \text{minimize}_{\mathbf{a},b,\boldsymbol{\delta},\boldsymbol{\sigma}} \quad & \textstyle\sum_{i=1}^{n} \delta_i + \sum_{j=1}^{m} \sigma_j \\ \text{subject to} \quad & \mathbf{x}_i^T \mathbf{a} + b + \delta_i \geq 1, \quad \forall i = 1, ..., n \\ & \mathbf{y}_j^T \mathbf{a} + b - \sigma_j \leq -1, \quad \forall j = 1, ..., m \\ & \delta_i \geq 0, \ \sigma_j \geq 0, \quad \forall i = 1, ..., n, \quad j = 1, ..., m \end{aligned}$$

Associate w_i to each of the first set of constraints, v_j to each of the second set of constraints. (Suppose $\mathbf{a}$ is a d-dimensional vector.)

Example Continued

The dual problem is

$$
\begin{aligned}
\text{maximize}_{\mathbf{w},\mathbf{v}} \quad & \sum_{i=1}^{n} w_i - \sum_{j=1}^{m} v_j & \\
\text{subject to} \quad & \sum_{i=1}^{n} x_{ik} w_i + \sum_{j=1}^{m} y_{jk} v_j = 0, & \forall k = 1, ..., d \\
& \sum_{i=1}^{n} w_i + \sum_{j=1}^{m} v_j = 0 & \\
& 0 \leq w_i \leq 1, & \forall i = 1, ..., n \\
& -1 \leq v_j \leq 0, & \forall j = 1, ..., m
\end{aligned}
$$

If we let $X = [\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_n]$ and $Y = [\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_m]$, then the first constraint can also be written as

$$X\mathbf{w} + Y\mathbf{v} = \mathbf{0}$$

Example Continued

The support vector machine problem:

$$\begin{aligned} \text{minimize} \quad & \textstyle\sum_i \delta_i + \sum_j \sigma_j \\ \text{subject to} \quad & \mathbf{x}_i^T \mathbf{a} + b + \delta_i \geq 1, \quad \forall i \\ & \mathbf{y}_j^T \mathbf{a} + b - \sigma_j \leq -1, \quad \forall j \\ & \delta_i \geq 0, \ \sigma_j \geq 0, \quad \forall i, j \end{aligned}$$

Dual:

$$\begin{aligned} \text{maximize} \quad & \textstyle\sum_i w_i - \sum_j v_j \\ \text{subject to} \quad & X\mathbf{w} + Y\mathbf{v} = \mathbf{0} \quad \text{— } x_i \ y_j \text{ are free} \\ & \textstyle\sum_i w_i + \sum_j v_j = 0 \quad \text{— } a \ b \text{ are free} \\ & 0 \leq w_i \leq 1, \quad \forall i = 1, \ldots, n \\ & -1 \leq v_j \leq 0, \quad \forall j = 1, \ldots, m \end{aligned}$$

Invariance of Transformations

Theorem

If we transform a linear program to an equivalent one (such as by replacing free variables, adding slack variables, etc), then the dual of the two problems will be equivalent.

Theorem

If we transform the primal to its dual, then transform the dual to its dual, then we will obtain a problem equivalent to the primal problem, that is, the dual of dual is the primal.

Duality Theorems

Remember that by the way we construct the dual problem, we argued that they should have the same optimal value.

- In the next lecture, we formally prove it.
- Since we have the invariant of transformation property, we only need to discuss the standard form. All the following theorems also apply to general cases as well.